

The Weighted Average Inventory (Stock Purchasing) Model.

Benefits of the Model

This framework provides a data-driven, systematic approach to inventory management. By balancing product velocity, profit margins, and cash liquidity, it removes the guesswork from purchasing decisions. The model ensures capital is not trapped in low-performing or "singular" items while protecting against stockouts of high-volume, highly profitable favorites.

How it Helps the Business Owner

For a business owner, cash flow is king. This framework helps owners protect their operational cash flow by integrating a mandatory "Financial Check" before any stock is ordered. It prevents overstocking by mathematically calculating exact re-order minimums based on actual sales and supplier lead times. Ultimately, it maximizes Return on Investment (ROI) by ensuring working capital is deployed only toward inventory that objectively performs well.

How it Helps the Advisor

For an advisor or consultant, this framework provides a structured, objective tool to evaluate a client's inventory health. It bridges the gap between operational metrics (like "days on hand") and financial metrics (like the "Quick Ratio"). This allows the advisor to elevate the conversation from simple supply-chain management to holistic business liquidity, risk assessment, and seasonal capital expenditure planning.

Who it is Suitable For

This model is highly suitable for retailers, wholesalers, e-commerce brands, and any inventory-heavy business. Because the framework includes strategic "exceptions" for different product types, it is particularly useful for businesses that carry a mix of high-volume commodities and unique, singular, or exotic items.

Using the Framework in Practice: Step-by-Step Methods

Step 1. Review Past Sales Reports & Collate Into Matrix

- The Action: Gather past sales data to determine which units yield the best margin, sell in the highest volumes, move the quickest, and carry the least unit cost risk.
- The Weighting: Assign a point weighting to each of these categories to generate a priority ranking or "Weighted Average" for your products.

- The Exceptions:
 - *For imported/exotic/singular items:* Ignore the 'Volume' weighting entirely. Place your emphasis on the 'Days on Hand' metric.
 - *For bulk/high-quantity items:* Ignore the 'Days on Hand' weighting. Heavily weight the 'Volume' sold (or sales as a % of stock units purchased), alongside 'Margin' and 'Unit Cost Risk'.

Step 2. Stock Check

- The Action: Compare your highest-ranking "Weighted Average" products against your actual physical stock holdings. Crucial: You must include incoming units currently in transit from suppliers.
- Calculate Re-order Minimum: Assess your average daily units sold against the supplier's lead time (time taken to arrive).
 - *Formula:* Daily Average Sales × Days to Arrive = Re-order Minimum.
 - *Example:* If you sell 3 items per day and it takes 5 days for the supplier to deliver, your absolute re-order minimum is 15 units.

Step 3. Financial Check

- The Action: Before getting the checkbook out, review the company's financial position.
- The Metrics: Calculate the Quick Ratio (Liquidity) to assess your total financial exposure.
- The Forecast: Cross-reference this stock purchase against the operational needs of the business, upcoming Capital Expenditure (Capex) forecasts, and your seasonally adjusted Sales & Expenses forecast.

Step 4. Order Stock

- The Action: Execute the purchase order confidently, knowing the stock is mathematically required and financially viable.

Common Q&A

Q1: What is the purpose of the Weighted Average?

A: It provides an objective, data-driven ranking of your inventory by synthesizing margin, volume, speed of sale, and unit cost risk into a single score.

Q2: How do I handle unique, high-value, or imported items in the matrix?

A: Because these items are "singular by nature," they won't have high volume. The framework instructs you to ignore the 'Volume' weighting and instead place heavy emphasis on the 'Days on Hand' figure.

Q3: How exactly do I calculate when to re-order?

A: Multiply your "average daily units sold" by the number of days the product takes to arrive from the supplier. This number becomes your re-order minimum.

Q4: Should I count stock that hasn't arrived yet?

A: Yes. When comparing your weighted average results against current stock, you must always include incoming units that are in transit.

Q5: Why is the Financial Check Step necessary if I know I need the stock?

A: Because operational needs and liquidity (Quick Ratio) take precedence. Even if stock is low, evaluating your cash position, upcoming Capex, and seasonal expense forecasts ensures you don't over-expose the business financially to buy inventory.

Q6: What if my sales are highly seasonal?

A: During Step 3, the framework requires you to view your Sales & Expenses forecast as "seasonally adjusted" so you aren't ordering based on peak-season daily averages during a slow period.

Sources Provided

Template Name	Core Focus & Contents
Inventory Framework	Explains a 4-step methodology for inventory management. Details how to create a "Weighted Average" product matrix using past sales data, how to calculate re-order minimums based on daily sales and lead times, and mandates a financial liquidity (Quick Ratio) check prior to ordering stock.

AI Logic Branch Chain: Inventory Assessment

Logic Branch	IF (Condition / Trigger)	THEN (Action / Next Step)	Additional Context / Notes
Matrix Exception: Exotic	IF the item is categorized as "singular" or "imported exotic"...	THEN ignore 'Volume' weighting and emphasize 'Days on Hand'.	Adjusts the model for high-value items that naturally move slower but carry high margins.
Matrix Exception: Bulk	IF the item is categorized as a "large quantity" stock item...	THEN ignore 'Days on Hand' weighting and emphasize 'Volume' sold and 'Unit Cost Risk'.	Ensures high-turnover commodities aren't penalized for high stock levels.
Re-Order Trigger	IF (Current Stock + In-Transit Stock) <= (Average Daily Units Sold × Days to Arrive)...	THEN trigger stock replenishment alert and move to Financial Check (Step 3).	Mathematically ensures stock won't run out before the supplier can deliver.
Financial Approval	IF the Quick Ratio is healthy AND available cash > upcoming Capex + seasonally adjusted operational expenses...	THEN proceed to Step 4: Order Stock.	Ensures the business has adequate liquidity to survive the cash conversion cycle of the new stock.
Financial Rejection	IF the Quick Ratio indicates low liquidity OR upcoming Capex requires the available cash...	THEN halt the order OR reduce order quantities to match strictly necessary daily minimums.	Prevents inventory purchases from bankrupting operational cash flow.